Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering

Batch: 20th
Date: 01/01/2025

Mid-Term Examination

Semester: Fall 2024
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any five questions: $5 \times 6 = 30$

1. (a) Illustrate different types of process flow. 3
   (b) What are the essences of software engineering practice? Explain. 3

2. (a) Write the five framework activities of Personal Software Project. 3
   (b) Describe three types of process pattern. 3

3. (a) List and explain all the phases of the Unified Process Model. 3
   (b) Explain evolutionary process models. 3

4. (a) What is Team Software Process? Explain. 3
   (b) What is Agility? List all Agility principles. 3

5. (a) Write the human factor in agile process. 3
   (b) Describe the XP concept of refactoring and pair programming in your own words. 3

6. (a) Define product backlog and burn down chart in Scrum. 3
   (b) List and explain a set of development actions in Scrum. 3

7. (c) What is a successful test in software construction? 3
   (d) How does agile communication differ from traditional software engineering communication? Explain. 3

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CSE423: Simulation and Modeling

Batch: 20th | Semester: Fall 2024

Date: 14-01-2025 | **Mid-term Examination** | Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings. $5 \times 8 = 40$

1. (a) Define any three terminologies of system simulation with example. 3
   (b) List any three situations when simulation tool is appropriate and when not appropriate. 3

2. (a) Briefly explain verification and validation in modelling. 3
   (b) Draw the flowchart of steps involved in simulation study. 3

3. (a) How does probability involved in discrete event simulation? 2
   (b) Sate the steps of discrete event simulation model. 4

4. (a) Explain the terms used in queuing notations of the form A/B/C/N/K. 2
   (b) List and define the primary performance measures of queueing systems. 4

5. (a) Explain poisson distribution in simulation study. 2
   (b) Draw the flow of control for the next event time advance approach. 4

6. (a) Differentiate between continuous and discrete systems. 2
   (b) Construct a diagram for arrival routine in queueing system. 4

7. (a) Define conceptual model with example. 2
   (b) Illustrate the four activities occur inside the single-server queueing system. 4

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CSE423: Simulation and Modeling

Batch: 20^{th} | Semester: Fall 2024

Date: 17-03-2025 | **Final Examination** | Time: 2 Hours | Total Marks: 40

Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) What are the advantages of using modeling and simulation for data analysis? 4
   (b) Why histogram is important in data distribution? How a histogram is constructed? 4

2. (a) What is Monte Carlo Simulation? What kind of input data is required for performing Monte Carlo simulations? 4
   (b) Write the general algorithm of Monte Carlo Simulation. 4

3. (a) When a stochastic process is called a Continuous-Time Markov Chain? 3
   (b) Explain Markov Process with an example. 5

4. (a) What are the key elements of a queuing system? 2
   (b) Create the events for aircraft arrivals, landing and departures queuing system. 6

5. (a) Explain the types of simulations with respect of output analysis with Example. 4
   (b) Draw the diagram for event scheduling scheme in computer simulation. 4

6. (a) List the issues that are considered in an accreditation decision 3
   (b) Explain inverse- transform technique of producing random variates for Exponential Distribution. 5

7. (a) Write short note validation in simulation. 2
   (b) Genarate Three poison variates with mean $\alpha = 0.2$. 6

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CSE 469 : Digital Image Processing

Batch: 20th | Semester: Fall 2024

Date: 20/03/2025 | **Final Examination** | Time: 2 Hours

Total Marks: 40

Answer any five questions. $5 \times 8 = 40$

1. (a) Define Image Segmentation. Mention different types of Image Segmentation. 3

   (b) State the steps for filtering in the frequency domain. 5

2. (a) Briefly describe different types of Data Redundancies in Images. 3

   (b) Determine the average length of the code from the following information after source reduction using the Huffman Coding algorithm. 5

Original Source	
Symbol	Probability
A1	0.1
A2	0.4
A3	0.06
A4	0.1
A5	0.04
A6	0.3

3. (a) Define Data Compression and Compression Ratio. 3

   (b) A computer-generated image has the intensity distribution shown in the second column of the following table. Now, determine the average number of bits – 5

   (i) if a natural 8-bit binary code (denoted as code 1 in the table) is used,

   (ii) if the scheme designated as code 2 in the table is used to represent its four possible intensities.

   For Code 2, find out the compression ratio.

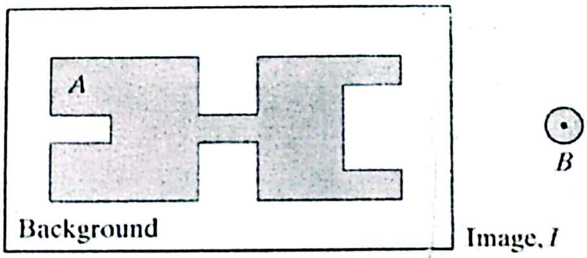
r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	01010111	8	1	1
$r_{186} = 186$	0.25	01010111	8	000	3
$r_{255} = 255$	0.03	01010111	8	001	3

4. (a) State Objective and Subjective Fidelity Criteria of assessment of information loss in images. 4

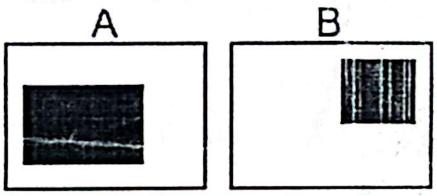
(b) Draw and describe the functional block diagram of a general Image Compression system. 4

5. (a) Define Dilation and Erosion. 2

(b) Consider the following image I with the structuring element B. Show the final effect after performing opening and closing operations on I using B. 6



6. (a) Illustrate the following logic operations on the given images A and B – (i) NOT, (ii) AND, (iii) OR, and (iv) XOR 4



(b) How does the size and shape of the structuring element (SE) affect the amount and manner in which an object thickens in morphological operations? Explain with an example. 4

7. (a) State the key conditions that must be satisfied when partitioning an image's spatial region R into subregions $R_1, R_2, \ldots, R_n$ during image segmentation. 2

(b) What are the convolution kernels used in the Sobel operator for edge detection? 2

(c) Describe and illustrate the steps for region splitting and merging for image segmentation. 4

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Batch: 20th
Date: 12/03/2025

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Answer any five questions: 5 x 8 = 40

1. (a) Develop a complete use case diagram for "Making a withdrawal from an ATM". 4
   (b) Write the steps to validate requirements. 4

2. (a) Explain the domain analysis in requirement modeling. 4
   (b) Describe the modeling types of requirement analysis. 4

3. (a) Draw a diagram for translating the Requirements Model into the Design Model. 4
   (b) Describe the Software Quality Guidelines in Design Process. 4

4. (a) How do we assess the quality of a software design? 4
   (b) Write the generic task set for design. 4

5. (a) Explain the golden rules for user interface design. 4
   (b) What is interface analysis? Explain. 4

6. (a) Illustrate a design pyramid for WebApp. 4
   (b) Why aesthetic design also called graphic design? Explain. 4

7. (a) Define Pragmatic View of software quality. 4
   (b) Explain the eight software design quality suggested by David Garvin. 4